

CASE 6

ADENOSINE DEAMINASE DEFICIENCY

The purine degradation pathway and lymphocytes.

In Case 5 we learned about severe combined immunodeficiency (SCID) and examined the most common form—that due to an X-linked mutation in the gamma chain (γ_c) common to several interleukin receptors. In that type of SCID, T cells are virtually absent, whereas B cells are present in normal numbers although they are not functional. Further examination of this phenotype also reveals an absence of natural killer (NK) cells. Thus, X-linked SCID is classified as T⁻B⁺NK⁻ SCID.

SCID patients with an almost complete absence of both B cells and T cells are also encountered; their phenotype is T⁻B⁻. This phenotype is exclusively associated with SCID transmitted as an autosomal recessive condition. Some forms of T⁻B⁻ SCID are due to defects of V(D)J recombination; these disorders have a normal number of NK lymphocytes (T⁻B⁻NK⁺ SCID). However, in other cases of T⁻B⁻ SCID, the number of NK lymphocytes is severely reduced (T⁻B⁻NK⁻ SCID). The most common genetic defect encountered in these patients is mutation in the gene encoding the purine-degradation enzyme adenosine deaminase (ADA). ADA is a ubiquitous housekeeping enzyme found in all mammalian cells; it converts the purine nucleosides adenosine and deoxyadenosine to inosine and deoxyinosine respectively, and hence to waste products that are excreted (Fig. 6.1). In its absence, cells can accumulate excessive amounts of phosphorylated adenosine and deoxyadenosine metabolites, which are toxic when present in excessive amounts. Lymphocytes are particularly susceptible to the toxic effects of these metabolites, and thus ADA

TOPICS BEARING ON THIS CASE:

Defects in lymphocyte function result in severe combined immunodeficiency

Bone marrow transplantation

Mixed lymphocyte reaction

Minor histocompatibility antigens

Graft-versus-host reaction

Case 6: Adenosine Deaminase Deficiency

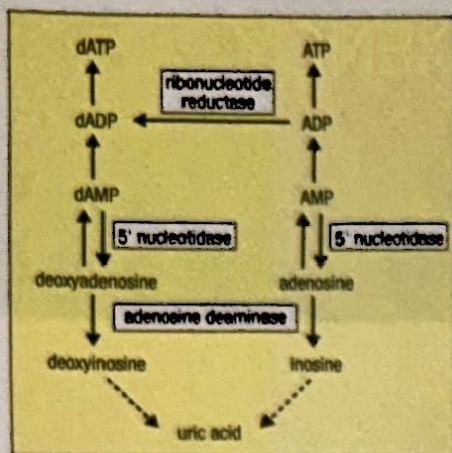


Fig. 6.1 Purine metabolic pathways. Adenosine deaminase catalyzes the conversion of deoxyadenosine and adenosine to deoxyinosine and inosine, respectively, which are eventually converted to waste products and excreted. In the absence of ADA in lymphocytes, adenosine and deoxyadenosine metabolites accumulate. In cells other than lymphocytes, the enzyme 5' nucleotidase can convert AMP and dAMP to adenosine and deoxyadenosine, thus preventing the build-up of potentially toxic metabolites.

3-week-old female infant with thrush. Family history of SCID.

Diagnosis of SCID. Consider bone marrow transplant.

Bone marrow transplant from HLA- and blood group-identical brother.

Fig. 6.2 Thymic shadow on chest radiographs of infants. In the left panel the thymic shadow in a normal healthy child is surrounded by a dashed blue line. In the right panel the heart shadow (surrounded by a dashed red line) is clearly visible in the SCID infant owing to the absence of the thymus. Courtesy of T. Griscom.

deficiency leads to SCID. However, because of the ubiquitous expression of the ADA gene, other cell types are also affected, accounting for the extraimmune manifestations of ADA deficiency.

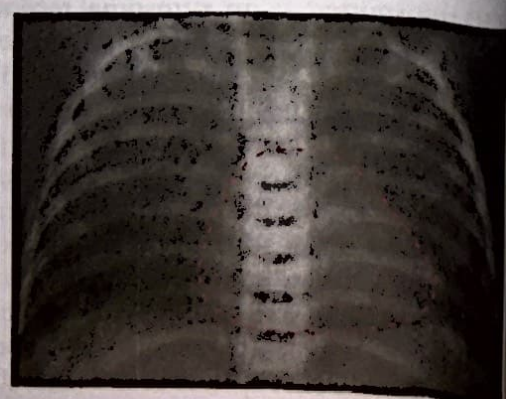
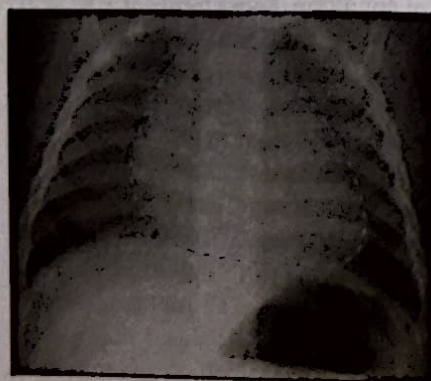
The case of Roberta Alden: lymphocytes poisoned by toxic metabolites.

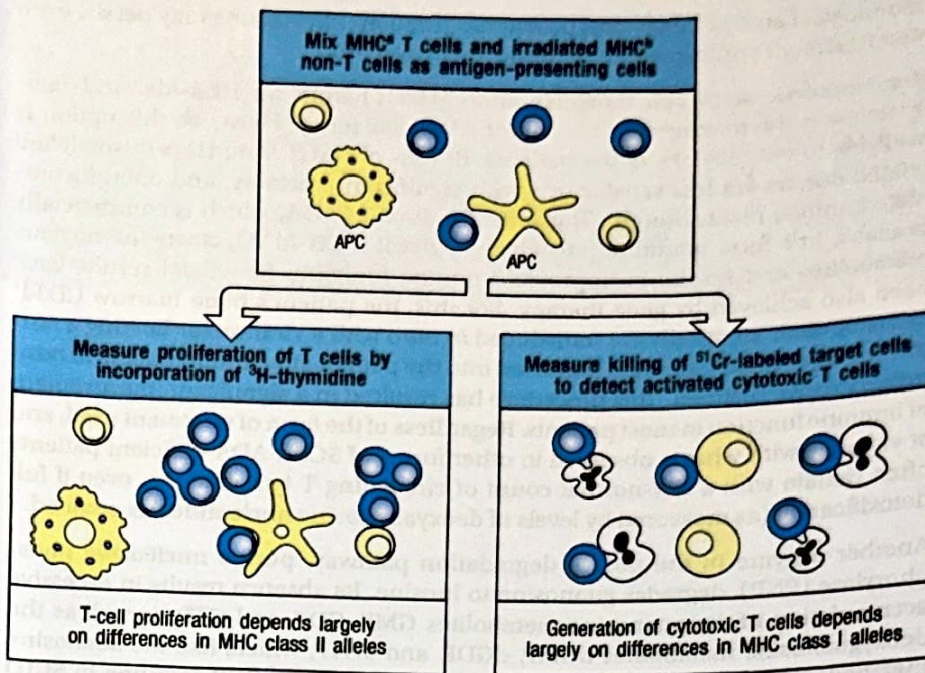
The Aldens are an African-American family from a remote rural area of the state of Georgia. Mr and Mrs Alden are probably distantly related to each other. At the time they moved to Boston they had seven healthy children—four boys and three girls. Their eighth child was a boy, who developed severe pneumonia at 3 months old and died at the City Hospital. An autopsy revealed that he had SCID. His thymus had a fetal appearance, with only rare thymocytes and no Hassall's corpuscles (see Fig. 5.2). Two years later the Aldens had a daughter, named Roberta. She appeared healthy at birth, but 6 weeks later Mrs Alden noticed thrush in Roberta's mouth (see Fig. 5.6).

Aware of the family history, Roberta's pediatrician ordered a chest X-ray and blood studies. No thymic shadow (Fig. 6.2) was seen in the chest X-ray, and the anterior margins of the ribs were flared. Roberta's lymphocyte count was $150 \text{ cells } \mu\text{l}^{-1}$ (normal for an infant is $>2500 \text{ cells } \mu\text{l}^{-1}$). Her lymphocytes did not respond to the non-specific T-cell mitogen phytohemagglutinin. A diagnosis of SCID was established.

HLA typing of Roberta, her parents, and her seven siblings revealed that her HLA type was identical with one sister, Ellen, and one brother, John. To confirm histocompatibility, when Ellen's and John's blood cells were tested separately against Roberta's in the mixed leukocyte reaction (Fig. 6.3), and no reaction was seen in either case. John and Roberta also had the same blood type (AB), so John was chosen as the bone marrow donor. Bone marrow cells (2.8×10^9 cells) were removed from John's iliac crest bone and infused into Roberta (this was calculated to be a dose of 5×10^6 cells kg^{-1} of Roberta's body weight).

Two months after the bone marrow transplant, Roberta's lymphocyte count had increased to $500 \text{ cells } \mu\text{l}^{-1}$, and the response of her blood lymphocytes to phytohemagglutinin had risen to half normal. A karyotype of the responding cells revealed that they had XY sex chromosomes; they were thus of male origin. Roberta continued to gain weight and was discharged from the hospital. Several weeks later she, her parents, and all her siblings were affected with a severe influenza-like respiratory infection. She recovered from this without problems and her lymphocyte count rose to $1575 \text{ cells } \mu\text{l}^{-1}$. Roberta continued to grow and develop normally and remained free of infections. All her lymphocytes continue to have an XY karyotype.





Adenosine deaminase deficiency.

The gene encoding adenosine deaminase (ADA) is located on chromosome 20. SCID due to ADA deficiency appears in individuals who carry on both copies of chromosome 20 a mutated variant of the ADA gene, resulting in an inactive enzyme or a lack of enzyme. Because production of ADA from the normal gene in heterozygotes is sufficient to compensate, the disease is inherited as an autosomal recessive condition (Fig. 6.4). Homozygotes or compound heterozygotes for mutations in this gene show the most profoundly lymphopenic form of SCID. The thymus is poorly developed and there is a characteristic abnormality in the rib bones.

Normally, the adenosine and deoxyadenosine content of cells is limited by ADA, which converts these nucleotides to inosine (and deoxyinosine) and subsequently to urate, which is excreted. The limited amounts of adenosine within cells are converted to adenosine monophosphate (AMP), adenosine diphosphate (ADP), and adenosine triphosphate (ATP). Deoxyadenosine is converted to dAMP, dADP, and dATP (see Fig. 6.1). In the absence of ADA, these metabolites can accumulate in up to 1000-fold excess within cells. Excess dATP in particular inhibits the enzyme ribonucleotide reductase, which is necessary for the synthesis of all the deoxynucleotides required for DNA synthesis (Fig. 6.5); its inhibition is probably the main culprit in causing the death and nondevelopment of lymphocytes.

The thymus contains 13 times more ADA than any other tissue in the body. Because adenosine metabolites are very toxic to lymphocytes, the high level of ADA in the thymus is probably crucial for normal thymocyte development. The reason that lymphocytes are particularly vulnerable to the accumulation of these metabolic poisons is probably because they are relatively deficient in the enzyme 5' nucleotidase. This enzyme degrades AMP and dAMP to adenosine and deoxyadenosine (see Fig. 6.1) and thereby prevents the excessive accumulation of ADP, ATP, dADP, and dATP, even in the absence of ADA.

However, because ADA is ubiquitously expressed, ADA deficiency is also characterized by extralymphatic manifestations. In addition to rib flaring and other bone changes, ADA-deficient patients often develop sensorineural deafness and

Fig. 6.3 The mixed leukocyte reaction (MLR) can be used to detect histoincompatibility. Circulating mononuclear cells (i.e., lymphocytes, monocytes, and dendritic cells) from the two individuals who are to be tested for compatibility are isolated from peripheral blood. The cells from one person (yellow, expressing MHC^B), which also contain antigen-presenting cells (APCs), are either irradiated or treated with mitomycin C; they will act as stimulator cells but cannot respond by DNA synthesis and cell division to antigenic stimulation by the other person's cells. The cells from the two individuals are then mixed (top panel). If the unirradiated leukocytes (the responders, blue, expressing MHC^A) contain alloreactive T cells, these will be stimulated to proliferate and differentiate to effector cells. Between 3 and 7 days after mixing, the culture is assessed for T-cell proliferation (bottom left panel), which is mainly the result of CD4⁺ T cells recognizing differences in MHC class II molecules, and for the generation of activated cytotoxic T cells (bottom right panel), which respond to differences in MHC class I molecules. When the MLR is used to select a bone marrow donor, the prospective donor's cells are used as the responder cells and the prospective recipient's cells as the stimulator cells.

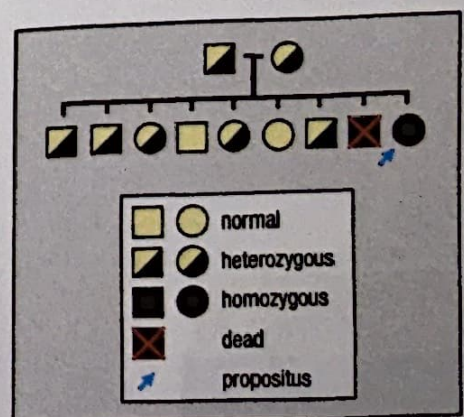


Fig. 6.4 Inheritance of ADA deficiency in Roberta's family. ADA deficiency is inherited as an autosomal recessive condition.

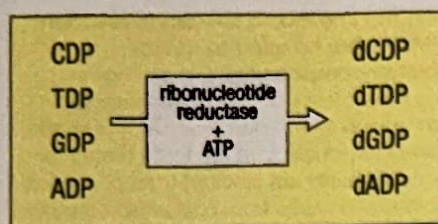


Fig. 6.5 Role of ribonucleotide reductase in generating deoxynucleotides required for DNA synthesis.

neurological and/or behavioral problems. These manifestations may persist even after treatment and successful immune reconstitution.

Hematopoietic stem cell transplantation (HSCT) from an HLA-identical family donor is the treatment of choice for ADA deficiency. However, this option is available to only 15–20% of the patients. Results of HSCT from HLA-mismatched related donors are less satisfactory, with significant mortality, and often incomplete immune reconstitution. The administration of ADA, which is commercially available in a form bound to polyethylene glycol (PEG-ADA), clears the noxious metabolites and results in improved immune function. Beneficial results have been also achieved by gene therapy. For this, the patient's bone marrow CD34⁺ hematopoietic stem cells are transduced *in vitro* with a viral vector bearing a normal ADA gene. They are then reinfused into the patient after a short course of non-myeloablative regimen. This procedure has resulted in a significant improvement of immune function in most patients. Regardless of the form of treatment used, and at variance with what is observed in other forms of SCID, ADA-deficient patients often remain with a low-normal count of circulating T lymphocytes, even if full detoxification (as measured by levels of deoxyadenosine nucleotides) is attained.

Another enzyme in the purine degradation pathway, purine nucleoside phosphorylase (PNP), degrades guanosine to inosine. Its absence results in excessive accumulation of the guanosine metabolites GMP, GDP, and GTP, as well as the deoxyguanosine metabolites dGMP, dGDP, and dGTP, which, like the adenosine metabolites, are toxic to lymphocytes. Genetic defects in PNP resulting in SCID have also been found. In most cases, there is progressive and severe T-cell lymphopenia, whereas the number of circulating B lymphocytes can be variable. In addition to severe infections, progressive neurological deterioration and autoimmune manifestations (in particular, autoimmune hemolytic anemia) are common. HSCT is the only curative treatment.

Questions.

- 1 The absence of a thymic shadow in the chest radiograph of a young infant can be helpful in the diagnosis of SCID, but it is not a reliable finding. Why is this so?
- 2 In a mixed lymphocyte reaction, the cells used to stimulate the response are treated with mitomycin, a mitotic inhibitor, to prevent their responding to the responder cells. When Roberta's and John's cells were tested in a mixed lymphocyte reaction to ascertain whether his T cells would respond to Roberta's cells, it was not necessary to treat Roberta's cells with mitomycin. Why was this the case?
- 3 Can you think of a reason for the fact that many patients with ADA deficiency fail to normalize their T-lymphocyte count, even after successful treatment and complete detoxification?